MATHEMATICAL PRELIMINARIES FOR MACHINE LEARNING

CS3244 Machine Learning

OUTLINE

01

Linear Algebra

02

Calculus

LINEAR ALGEBRA

WHY LINEAR ALGEBRA?

- Geometric interpretation for distance
- ML involves the study of multivariate functions!
- Simplest models relies on Linear Algebra
- More complicated models combines linear models in a non-linear manner
- Applications in multivariate calculus

WHAT YOU SHOULD KNOW?

- Vectors & Vector Spaces
- Subspaces
- Dot Product
- Linear Map and Matrices

VECTOR SPACES

- Vector spaces are useful mathematical structures
- A collection of objects, vectors, that satisfies some mathematical axioms
- Most commonly encountered setting in ML is Euclidean Space or $\mathbb{R}^n$
- Can decompose each vector in to sum of its basis vectors (largest linearly independent set)

VECTORS

Concretely what is a vector? It is just an n – tuple of real numbers:

$\boldsymbol{u} = (u_1, u_2, \dots, u_n) \in \mathbb{R}^n$ (boldface to denote vector)

$\forall\, i = 1, 2, \dots, n,\ u_i \in \mathbb{R}$ (scalar not bolded)

(each component is a real number – subscript denotes positional index)

(*n* of such scalars because *n* – dimensional vector)

RECAP OF VECTOR OPERATIONS

Let $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$ be n – dimensional real vectors and $x \in \mathbb{R}$

Adding two vectors:

$$\boldsymbol{u} + \boldsymbol{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$

Addition is associative (informally, the order in which addition is done for more than 2 vectors does not matter):

$$\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$$

Addition is commutative: (adding $\boldsymbol{u}$ to $\boldsymbol{v}$ is the same as adding $\boldsymbol{v}$ to $\boldsymbol{u}$)

$$\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$$

RECAP OF VECTOR OPERATIONS

Scalar Multiplication (done component – wise):

$$x\boldsymbol{u} = (\, xu_1, xu_2, \ldots, xu_n)$$

Distributivity of Multiplication (multiply each component of sum is same as multiplying the sum):

$$x(\boldsymbol{u} + \boldsymbol{v}) = x\boldsymbol{u} + x\boldsymbol{v}$$

SUBSPACES

Intuitively – subspace is a smaller vector space carved out from bigger / ambient vector space

$S \subseteq \mathbb{R}^n$ is a subspace if it satisfies the following:

1. $\mathbf{0} \in S$, where $\mathbf{0} = (0, 0, \ldots, 0)$
2. For any vector $\boldsymbol{u} \in S$ **and** scalar $x \in \mathbb{R}$, $x\boldsymbol{u} \in S$
3. For any vector $\boldsymbol{u}, \boldsymbol{v} \in S$, $\boldsymbol{u} + \boldsymbol{v} \in S$

EXAMPLES:

1. Trivial Subspace, $\{\mathbf{0}\}$, space containing only the 0 – vector is a subspace
   Sometimes also known as the null-space.
2. $\{(x, y, 0) \mid x, y \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$
3. $\{(x, y, z, w) \mid x, y, z, w \in \mathbb{R}, x + y + z + w = 0\}$ is a subspace of $\mathbb{R}^4$

   In fact this is a 3 – dimensional subspace. Any subspace with 1 dimension less than ambient space is called a hyperplane.

NON-EXAMPLES

1. $\{(x, 0) \mid x \in \mathbb{R}\} \cup \{(0, y) | y \in \mathbb{R}\}$ is NOT a subspace of $\mathbb{R}^2$ (not closed under addition)
2. $\{(x, y, 1) | x, y \in \mathbb{R}\}$ is NOT a subspace of $\mathbb{R}^3$, but it is an affine hyperplane (does not contain $\mathbf{0}$)
3. $\{(x^2, 0, 0) \mid x \in \mathbb{R}\}$ is NOT a subspace of $\mathbb{R}^3$ (not closed under scalar multiplication)

WHY STUDY SUBSPACES

01

Hyperplanes are useful in classification problems

02

Complicated classification models combine hyperplanes in a non-linear manner

03

Important in dimensional reduction (we project to a lower dimensional subspace)

DOT PRODUCT

Dot product: denoted $\boldsymbol{u}^\top \boldsymbol{v}$ (think Matrix Multiplication!)

(Notation used to prevent confusing dots used in scalar multiplication)

Sometimes also called Scalar Product

Take summing up the component-wise product:

$$\boldsymbol{u}^\top \boldsymbol{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n = \sum_{i=1}^{n} u_i v_i$$

WHY DOT PRODUCT?

01

Dot product induces a norm (a notion of length, distance) in vector space.

02

Gives notion of angle and geometry between vectors (useful in higher dimensional space)

03

Can generate orthonormal basis – whose vectors are "perpendicular"

04

Generalisation of dot product – Inner Product! Useful for Support Vector Machines

PROPERTIES OF DOT PRODUCT

Here is a recap of properties of Dot Products

Commutativity:

$$\boldsymbol{u}^\top \boldsymbol{v} = \boldsymbol{v}^\top \boldsymbol{u}$$

Compatible with Scalar Multiplication:

$$(x\boldsymbol{u})^\top \boldsymbol{v} = x(\boldsymbol{u}^\top \boldsymbol{v}) = \boldsymbol{u}^\top (x\boldsymbol{v})$$

Cauchy – Schwarz Inequality:

$$|\boldsymbol{u}^\top \boldsymbol{v}| \leq \|\boldsymbol{u}\|_2 \, \|\boldsymbol{v}\|_2$$

Induces a Norm:

$$\boldsymbol{u}^\top \boldsymbol{u} = \|\boldsymbol{u}\|_2^2$$

NORMS:

Norm, $\|\cdot\|$, encodes the idea of distance and length. This idea is useful in ML in defining closeness and distance

Positive Definite: $\|\boldsymbol{u}\| \geq 0$

Respects Scalar Multiplication: $\|x\boldsymbol{u}\| = |x|\|\boldsymbol{u}\|$

Satisfies Triangle Inequality: $\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$

EXAMPLES OF NORMS:

l2 Norms: $\|\boldsymbol{u}\|_2 = (\sum_{i=1}^{n} |u_i|^2)^{\frac{1}{2}}$

l1 Norms: $\|\boldsymbol{u}\|_1 = \sum_{i=1}^{n} |u_i|$

Sup Norm: $\|\boldsymbol{u}\|_\infty = \lim_{p\to\infty} (\sum_{i=1}^{n} |u_i|^p)^{\frac{1}{p}} = \max\{u_1, u_2, \dots, u_n\}$

Norms of functions: If $|f(x)|^2$ is an integrable function over entire reals, then we can define L2 Norm: $\|f(x)\|_2 = \left(\int_{-\infty}^{\infty} |f(x)|^2 \, dx\right)^{\frac{1}{2}}$

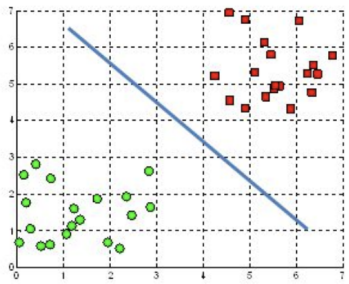
DESCRIBING A HYPERPLANE

How to describe points on a hyperplane?

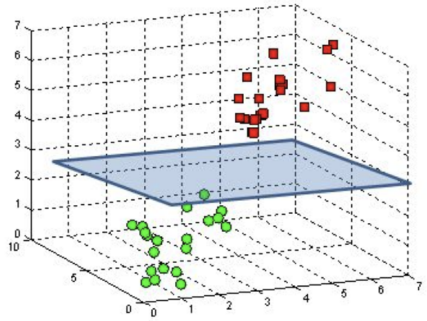
You can use dot product to describe all the points in the hypersurface!

If you can do this by finding out the perpendicular vector of the hyperplane!

A hyperplane in $\mathbb{R}^2$ is a line

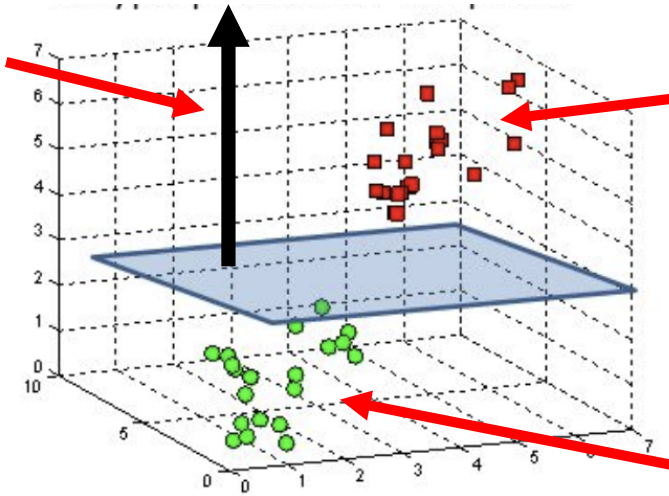


A hyperplane in $\mathbb{R}^3$ is a plane



IN A NUTSHELL : LINEAR CLASSIFIERS

Vector orthogonal to the plane



Positively Classified Examples
(above the plane, having positive dot product)

Negatively Classified Examples
(below the plane, having negative dot product)

IN A NUTSHELL : LINEAR CLASSIFIERS

Let $\boldsymbol{u}$ be the perpendicular vector of the hyperplane. Any two point (vector, $\boldsymbol{v_1}, \boldsymbol{v_2}$) on the plane satisfies:

$$(\boldsymbol{v_1} - \boldsymbol{v_2})^\top \boldsymbol{u} = 0$$

Let $x_0 = -\boldsymbol{v_1}^\top \boldsymbol{u}$

So for any point on plane:

$$\boldsymbol{v}^\top \boldsymbol{u} + x_0 = 0$$

If $\boldsymbol{w} = (w_1, \dots, w_n)$ is a point that does not fall on the plane,

either $\boldsymbol{w}^\top \boldsymbol{u} > 0$ (positively classified) or $\boldsymbol{w}^\top \boldsymbol{u} < 0$ (negatively classified)

How to train? More details in CS3244

LINEAR MAP

Linear Maps are functions that preserves linearity of operations between two vector spaces. Let V, W be real vector spaces and $T: V \to W$ is a function from V to W.

T is called a linear transformation if it satisfies the following:

For any vectors $\boldsymbol{v_1}, \boldsymbol{v_2} \in V$ and any scalar $x \in \mathbb{R}$, we have that:

$$T(\boldsymbol{v}_1 + \boldsymbol{v}_2) = T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2)$$

$$T(x\boldsymbol{v}_1) = xT(\boldsymbol{v}_1)$$

We can talk about linear maps in the abstract but concretely, in finite dimensional vector space, matrices encode behaviour of linear maps.

WHY STUDY LINEAR MAPS?

1. Appears in Dimensionality Reduction (we change basis)
2. Mapping respects “linearity”
3. A way to understand differentiation (which we will see later)

EXAMPLES OF LINEAR MAPS

- Multiplication map: $f_a(x) = ax$
- Matrices: Let $\boldsymbol{M}$ be a $m \times n$ matrix then let $T_{\boldsymbol{M}}: \mathbb{R}^n \to \mathbb{R}^m$ with $T_{\boldsymbol{M}}(\boldsymbol{v}) = \boldsymbol{M}\boldsymbol{v}$
- Dot Product: Just a special case of matrix multiplication!
- A linear map that sends a vector to $\mathbb{R}$ is called a linear functional

OTHER COOL TOPICS NOT COVERED HERE:

Spectral theorems – Eigenvalue, Spectral decomposition and many other cool stuff

General Inner Product Space – Shows up in SVM

Some other resource you might want to refer to:

3Blue1Brown – Very complete and resourceful linear algebra content

Very resourceful LA content from Imperial College London:
https://www.youtube.com/watch?v=T3TpdPmTLso, part 1 of 3 of the playlist here:

https://www.youtube.com/playlist?list=PLmAuaUS7wSOP-iTNDivR0ANKuTUhEzMe4
featuring Multivariate Calculus and Principal Component Analysis (an important dimensionality reduction technique)

CALCULUS

WHY CALCULUS?

- A tool for Optimisation
- General Heuristic in Supervised Learning:
- Given training data, what are the parameters that will give me smallest amount of error? (Cost)
- Given training data, what are the parameters that will maximise my degree of accuracy? (Utility)

WHAT YOU SHOULD KNOW?

- Derivatives - in $\mathbb{R}^n$
- How to differentiate?
- Chain Rule
- Taylor Polynomials

WHAT IS DIFFERENTIATION

Let's recall the definitions of derivative in for real – valued function $f: \mathbb{R} \to \mathbb{R}$ in 1 dimensional real line. The derivative of f at a, denoted by $f'(a)$ is defined as

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

Or equivalently:

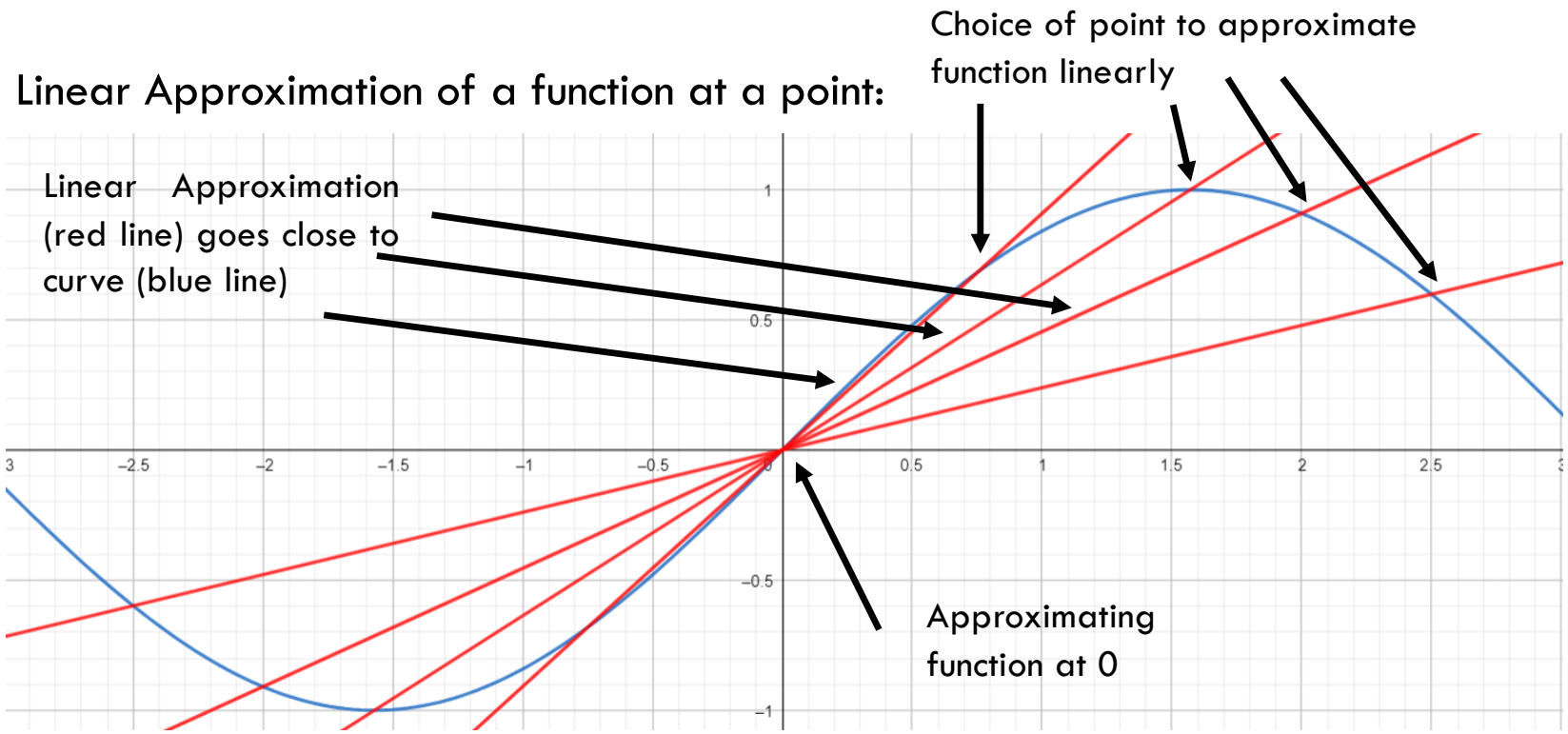
$$\lim_{h \to 0} |f(a+h) - f(a) - f'(a)(h)| = 0$$

But what exactly are we doing here?

We are taking the best linear approximation of the function at a point

WHAT IS DIFFERENTIATION?

Linear Approximation of a function at a point:



WHAT IS DIFFERENTIATION?

Intuitively, line gets better and better at approximating change in function near point at 0.

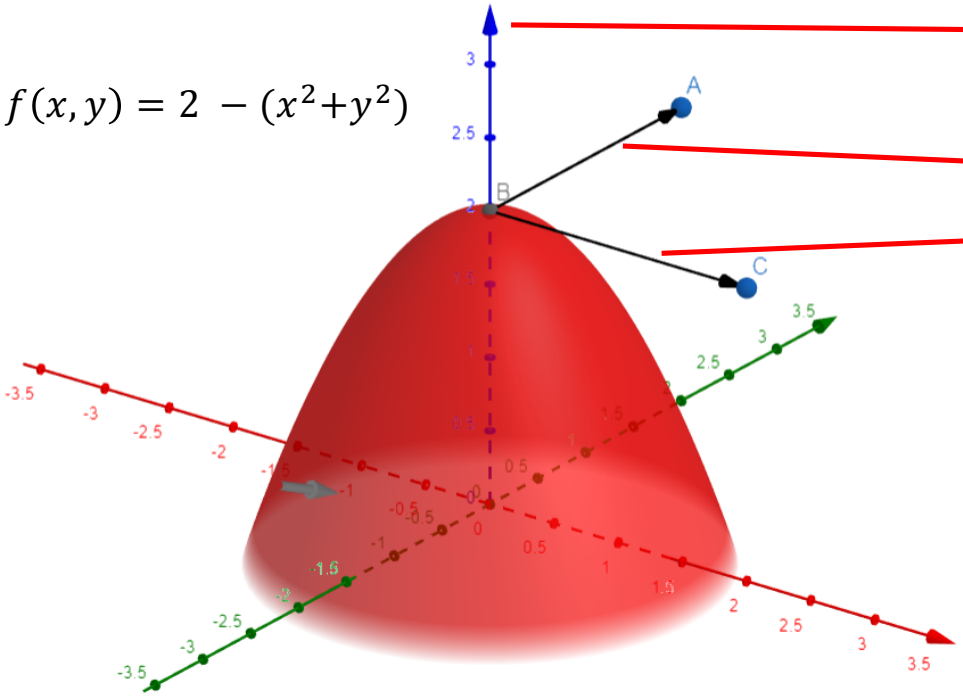
Differentiation is just the "best" linear approximation!

$$|\underbrace{f(a+h)-f(a)}_{\substack{\text{Difference in}\\ \text{values of } f \text{ near } a}} - \underbrace{f'(a)(a+h-a)}_{\substack{\text{Linear difference}\\ \text{between } a+h \text{ and } a\\ \text{with a linear factor}}}| < \underbrace{\varepsilon}_{\substack{\text{Distance can become}\\ \text{arbitrarily small}}}$$

How to generalise this higher dimensional space? – Linear Maps!

WHAT IS DIFFERENTIATION ? (VECTOR VALUED FUNCTION)

$f(x, y) = 2 - (x^2 + y^2)$



View z – axis as the value function takes

"small change" in z when we take a small step in x, y direction – approximates the function at point

WHAT IS DIFFERENTIATION? (FROM $\mathbb{R}^n \to \mathbb{R}$)

We replace a with $\boldsymbol{a} = (a_1, \dots, a_n)$, x with $\boldsymbol{x} = (x_1, \dots, x_n)$ and

$\nabla f(\boldsymbol{a}) = (f_1{}'(a_1),\ \ f_2{}'(a_2), \dots, f_n{}'(a_n))$ - for some suitable functions $f_i{}'(x)$ Natural, for $\nabla f(\boldsymbol{a})$ to be a vector:

$$|\underbrace{f(\boldsymbol{x}) - f(\boldsymbol{a})}_{\substack{\text{Difference in} \\ \text{values of } f \text{ near } \boldsymbol{a}}} - \underbrace{\nabla f(\boldsymbol{a})^\top (\boldsymbol{x} - \boldsymbol{a})}_{\substack{\text{Linear difference} \\ \text{between } \boldsymbol{x} \text{ and } \boldsymbol{a} \text{ with} \\ \text{some linear factor}}}| < \varepsilon$$

How to calculate the functions $f_i{}'(a_1)$?

HOW TO CALCULATE DERIVATIVES:

If $f(\boldsymbol{a}) = f(a_1, a_2, \dots, a_m)$, the partial derivatives at $\boldsymbol{a}$ with respect to variable x_i (i-th component) can be regarded like the single variable case:

$$\frac{\partial f}{\partial x_i}(\boldsymbol{a}) = \lim_{h \to 0} \frac{f(\boldsymbol{a} + h \cdot \boldsymbol{e}_i) - f(\boldsymbol{a})}{h}$$

where $\boldsymbol{e}_i = (0, \dots, 1, \dots 0)$ has 1 in the i -th component and 0 everywhere else

($\boldsymbol{e}_i$ is also called the "standard" basis)

We consider the derivative component-wise: Easy to see that: $\nabla f(\boldsymbol{a})_i = \frac{\partial f}{\partial x_i}(\boldsymbol{a})$.

In fact: $\nabla f(\boldsymbol{a}) = \left(\frac{\partial f}{\partial x_1}(\boldsymbol{a}), \frac{\partial f}{\partial x_2}(\boldsymbol{a}), \dots, \frac{\partial f}{\partial x_n}(\boldsymbol{a}) \right)$ except for some deviant cases which is not the main focus of the discussion here.

CHAIN RULE?

No problem for partial derivatives, in fact works just like single-variable case:

Single Variable Case:

$$h(x) = f\big(g(x)\big)$$
$$h'(x) = g'(x)f'(g(x))$$

Similarly, chain rule for partial derivatives, since we can view each component as a function from $g_i: \mathbb{R} \to \mathbb{R}$ by fixing other components, the usual chain rule applies:

Let $g\colon\ \mathbb{R}^m \to \mathbb{R},\ f: \mathbb{R} \to \mathbb{R}$ such that $g, f \in C^1$ once-continuously differentiable:

$$h(\boldsymbol{x}) = f\big(g(\boldsymbol{x})\big)$$

$$\frac{\partial h}{\partial x_i} = \frac{\partial f}{\partial g}(\boldsymbol{a})\frac{\partial g}{\partial x_i}(\boldsymbol{a}),$$

SOME EXAMPLES

1. $f(x_0, x_1, x_2) = x_0 + \ a_1 x_1 + a_2 x_2 \rightarrow \nabla f = (1, a_1, a_2)^\top$
   This says that the linear map is its own best approximation!
2. $h(x_0, x_1, x_2) = \exp(x_0 + a_1 x_1 + a_2 x_2)$ use chain rule:
   we see that : $h(x_0, x_1, x_2) = \exp(f)$

$$\frac{\partial h}{\partial f} = \ \exp(s)\,, \frac{\partial f}{\partial x_0} = 1, \frac{\partial f}{\partial x_1} = a_1, \frac{\partial f}{\partial x_1} = a_2 \rightarrow$$
$$\nabla h = \exp(x_0 + a_1 x_1 + a_2 x_2)\,(1\,, a_1\,, a_2)^\top$$

3. Left as an exercise: $\sigma(x_0, x_1, x_2) = (1 + h)^{-1}$
   σ is also the sigmoid function! An important function that appears in learning problems!

WHAT IS DIFFERENTIATION? (FROM $\mathbb{R}^n \to \mathbb{R}^m$)

Using similar ideas from the previous slide, we can extend the idea of differentiation to vector-valued functions $f(\boldsymbol{x}): \mathbb{R}^n \to \mathbb{R}^m$

In this case we can split the image of f up into m components and view the derivative component by component: so say $f_i(\boldsymbol{x}): \mathbb{R}^n \to \mathbb{R}, \boldsymbol{x} \to f(\boldsymbol{x})_i$ the i -th component of the vector $f(\boldsymbol{x})$ Observe that each $\nabla f_i(\boldsymbol{x}) \in \mathbb{R}^n$ is a vector:

$$\nabla f_i(\boldsymbol{x})^\top = \left(\frac{\partial f_i}{\partial x_1}, \frac{\partial f_i}{\partial x_2}, \dots, \frac{\partial f_i}{\partial x_n}\right)$$

We also know that this is the best "linear approximation" for *i*-th component of f

WHAT IS DIFFERENTIATION? (FROM $\mathbb{R}^n \to \mathbb{R}^m$)

Naturally, we can consider a $m \times n$ matrix $\boldsymbol{D}_f$ or $\boldsymbol{J}_f$ sometimes, $\boldsymbol{J}$ (called the Jacobian) such that the i-th row is $\nabla f_i(\boldsymbol{x})^\top$.

$$\mathbf{J} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

i-th row is $\nabla f_i(\boldsymbol{x})^\top$

Intuitively, we know that such a matrix is a the best linear map because i-th row is best linear approximation for i-th component, so this matrix approximates f well for nearby points.

Why is this important – used extensively in neural networks!

TAYLOR'S THEOREM – SINGLE DIMENSION

Recall in single variable calculus we have the Taylor's theorem:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)(x-a)^2}{2!} + \ldots + \frac{f^{(k)}(a)(x-a)^k}{k!} + \cdots$$

For values of $f(x)$ near $f(a)$.

Intuitively, if we were to approximate f near a using degree 1 polynomial (linearly), we know that the error term is controlled by a $(x - a)^2$

We have a similar result in multidimensional calculus!

TAYLOR'S THEOREM – MULTIPLE DIMENSIONS

If $f\colon \mathbb{R}^n \to \mathbb{R}$, and is differentiable at $\boldsymbol{a} = (a_1, a_2, \dots, a_n)$then near a, f will have the following multivariate polynomial expansion

$$f(\boldsymbol{x}) = f(\boldsymbol{a}) + \nabla f(\boldsymbol{a})^\top(\boldsymbol{x} - \boldsymbol{a}) + (\boldsymbol{x} - \boldsymbol{a})^\top \nabla^2 f(\boldsymbol{a})(\boldsymbol{x} - \boldsymbol{a}) + \cdots$$

Where $\nabla^2 f(\boldsymbol{a})$ (the Hessian) is a $m \times m$ matrix satisfying: $\nabla^2 f(\boldsymbol{a})_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{a})$

and can be generalised to higher derivatives; also equivalently:

$$f(\boldsymbol{x}) = f(\boldsymbol{a}) + \nabla f(\boldsymbol{a})(\boldsymbol{x} - \boldsymbol{a}) + O(|\boldsymbol{x} - \boldsymbol{a}|^2)$$

this is true because we expect $\nabla^2 f(\boldsymbol{a})$, for any vector $\boldsymbol{v}$, $|\nabla^2 f(\boldsymbol{a})\boldsymbol{v}| \leq c|\boldsymbol{v}|$ which is true in finite dimensional case

INTRODUCTION TO GRADIENT DESCENT

- Functions encountered in Machine Learning are nice enough (in the sense that the are differentiable almost everywhere), but still difficult to find closed form for minimum point.
- Use numerical methods to find the minima
- Gradient descent! – as a naïve heuristic, we take a small "step" that will give us the greatest amount of ascent/descent.

$$f(\boldsymbol{x}_{k+1}) = f(\boldsymbol{x}_k) + \nabla f(\boldsymbol{x}_k)(\boldsymbol{x}_{k+1} - \boldsymbol{x}_k) + O(|\boldsymbol{x}_k - \boldsymbol{x}_{k+1}|^2)$$

Intuitively, ideal choice for $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \eta \nabla f(\boldsymbol{x}_k)$ for some sufficiently small η:

$O(|\boldsymbol{x}_k - \boldsymbol{x}_{k+1}|^2)$ controlled by η^2 while $\nabla f(\boldsymbol{x}_k)(\boldsymbol{x}_{k+1} - \boldsymbol{x}_k) = -\eta \|\nabla f(\boldsymbol{x}_k)\|^2$

Don't worry if you don't get this. We'll cover it again later!

OTHER AWESOME MATERIALS REVOLVING MULTIVARIATE CALCULUS:

- Khan Academy on Youtube – Wide range of resources
- https://www.youtube.com/watch?v=tlpKfDc295M&list=PLSQl0a2vh4HC5feHa6Rc5c0wbRTx56nF7&index=19 – Video on gradients
- https://www.youtube.com/watch?v=bohL918kXQk&list=PLSQl0a2vh4HC5feHa6Rc5c0wbRTx56nF7&index=71– Video on Jacobian
- https://www.youtube.com/watch?v=VmfTXVG9S0U&list=PLSQl0a2vh4HC5feHa6Rc5c0wbRTx56nF7&index=69 – Some visuals and intuition on how to view higher dimensional derivatives as linear maps
- https://explained.ai/matrix-calculus/index.html: Explained AI, a webpage offering explanation of Machine Learning related topics. This posts goes more in-depth into Jacobian and with some worked examples from Neural Networks!
- https://www.youtube.com/watch?v=m998PdOCFcY&list=PLmAuaUS7wSOP-iTNDivR0ANKuTUhEzMe4&index=3&t=976s – from Imperial College London, part 2 of 3 of the playlist mentioned earlier.